

MIXING CHAMBER

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ABSTRACT

In this report, I will analyze the mixing chamber CFD analyze. It will be transient solution. Imagine a mixing chamber which placed inside a power plant. The mixing chamber has two inlet. One of them is cold fluid inlet. The other one is hot fluid inlet. Fully closed valve placed before the fluid inlet. At the beginning of the analyze it will be closed and we will open this valve immediately. I will analyze to what happens inside the chamber fluid behavior for first sec. I will add some blockage inside the chamber so fluid comes to the chamber and hit the blockage after that turn the flow way between the blockage and it will be reach to the outlet. Our objective is the determining the fluid temperature for the first 5 sec.

NOMENCLATURE

$\dot{m}$	mass flow rate
E	Energy
W	Work
Q	Heat
V	Speed
A	Cross sectional area
ρ	Density
h	Enthalpy

1. INTRODUCTION

Investigation of Swirling Flows in Mixing Chambers

This investigation analyzed the three-dimensional momentum and mass transfer characteristics arising from multiple inlets and a single outlet in micro mixing chamber. The chamber consists of a right square prism, an octagonal prism, or a cylinder. extensional local flows which produce steady stirring, and the configuration of colored particles at the outlet section expressed at different Re represented the mixing performance qualitatively. The combination of the Taylor dispersion and the vorticity was first introduced and made the mixing successful. The effects of various geometric parameters and Reynolds numbers on the mixing characteristics were investigated. An optimal design of the cylindrical chamber with 4 inlets can be found. At larger Reynolds number, $Re > 15$, more inertia caused the powerful swirling flows in the chamber, and

more damping effect on diffusion was diminished, which then increased the mixing performance.[1]

A phase guided passive batch microfluidic mixing chamber for isothermal amplification

With a view to developing a rapid pathogen detection system utilizing isothermal nucleic acid amplification, the necessary micro-mixing step is innovatively implemented on a chip. Passive laminar flow mixing of two 6.5 ml batches differing in viscosity is performed within a microfluidic chamber. This is achieved with a novel chip space-saving phase guide design which allows, for the first time, the complete integration of a passive mixing structure into a target chamber. Sequential filling of batches prior to mixing is demonstrated. Simulation predicts a reduction of diffusive mixing time from hours down to one minute. A simple and low-cost fabrication method is used which combines dry film resist technology and direct wafer bonding. Finally, an isothermal nucleic acid detection assay is successfully implemented where fluorescence results are measured directly from the chip after a one minute mixing sequence. In combination with our previous work, this opens up the way towards a fully integrated pathogen detection system in a lab-on-a-chip format.[2]

Design, construction, and preliminary testing of a square cross-sectional mixing chamber

This particular project aimed at designing and constructing a test chamber to provide gas mixtures in preparation for the CERT Tests. Preliminary tests were conducted to validate the setup. The chamber is made of plywood and clear polycarbonate, powered by a small 2 hp fan used to push the air through the chamber to the room. Based on previous work, High Efficiency Vortex (HEV) mixer design has been adapted to mix test gas with inlet air. The HEV mixer is placed directly after the fan and uses two sets of inline baffles to produce turbulent mixing. The testing included the delivery of the gas (fog) to the chamber, the determination of the optimal setup for all experiments, and the preliminary videos. An optimal height for the inlet nozzle of five inches was determined and the similarity between the fog speed and the air speed on the lowest fog and fan setting has been proven. Finally, the preliminary videos of the chamber flow clearly show a large scale vertical structure which is conducted along the length of the test section. This undesirable structure is caused by the fan. A flow

straightener could be used to reduce or eliminate this phenomenon.[3]



Figure 1: Fog to chamber setup

2. MATERIALS AND METHODS

In the setup part, at the general window important thing is time. It will be transient solution because we will like to see what happened to first 5 secs when we open the valve.

It is a heat transfer problem. There is a cold fluid and hot fluid. So we should turn on the energy equation and our flow is the turbulent so I open the k-epsilon turbulence model to analyze.

Transient solution:

Transient solution is time vary flow and heat transfer but for example, there is a vortex. If we analyze this simulation. We can see how these vortex formed from the beginning and moving forward to the outlet in the time. It is the good way to understand fluid behavior in the time. Time parameters which are time step size and number of time step are important for to analyze. Because the transient solving is the iterative implicit processes and solves first the first time step and it is iterative process and repeat under the reach total time of the analyze. Because it is not time dependent problem. Total time is multiply time step size and number of time step. If your number of time step is infinity, total time is infinity so problem is steady state.

$$[total\ time] = [time\ step\ size] * [number\ of\ time\ steps]$$

$$5\text{sec} = 0.005 * 1000$$

2.1. THE FIRST LAW OF THERMODYNAMICS:

The first law of thermodynamics, also known as the conservation of energy principle, provides a sound basis for studying the relationships among the various forms of energy and energy interactions. Based on experimental observations, the first law of thermodynamics states that energy can be neither created nor destroyed during a process; it can only change forms.[4]

ENERGY BALANCE:

$$\Delta E_{in} - \Delta E_{out} = E_{sys}$$

ENERGY ANALYSIS OF UNSTEADY-FLOW PROCESSES:

The energy content of a control volume changes with time during an unsteady-flow process. The magnitude of change depends on the amount of energy transfer across the system boundaries as heat and work as well as on the amount of energy transported into and out of the control volume by mass during the process. When analyzing an unsteady-flow process, we must keep track of the energy content of the control volume as well as the energies of the incoming and outgoing flow streams. Unsteady-flow, or transient flow, processes involve changes within the control volume with time.

The mass balance for any system undergoing any process can be expressed as (mentioned previously):

$$m_{in} - m_{out} = \Delta m_{cv}$$

Then the energy balance for a uniform-flow system can be expressed explicitly as

$$Q_{in} + W_{in} + \sum_{in} m\theta - Q_{out} + W_{out} + \sum_{out} m\theta = (m_2 e_2 - m_1 e_1)_{system}$$

Mixing chamber:

The kinetic and potential energies of the fluid streams are usually negligible ($ke = 0$, $pe = 0$). Then all there is left in the energy equation is the total energies of the incoming streams and the outgoing mixture, or

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 - \dot{m}_3 h_3 = 0$$

Calculation:

$$\dot{m} = \rho v A = 998.2 * 0.06 * 0.05 * 0.025 = 0.074865\text{ kg/s}$$

Eddy:

The movement of the fluid particle that is transported from one layer to another during the momentum exchange from one fluid layer to another in turbulent flow is known as "eddy". This momentum exchange in turbulent flow causes the velocity distribution to be more uniform than laminar flow as it moves away from the wall.

3. RESULTS AND DISCUSSION

Mixing chamber case 1 and case2

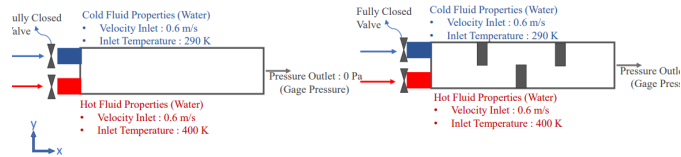


Figure2 : mixing chamber case1 and 2 properties

Residuals and temperatures for case-1:

- Flow time = 5s, time step = 1000

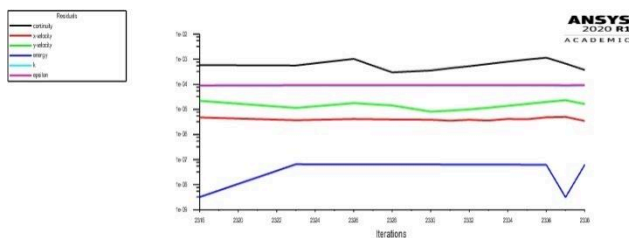


Figure3: residual for 5s case1

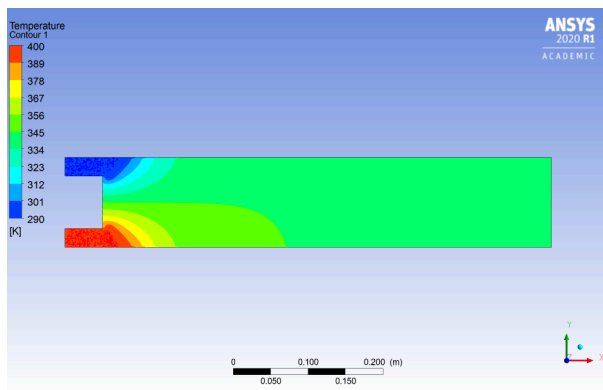


Figure4: temperature contour for 5s case1

- Flow time = 10s, time step = 2000

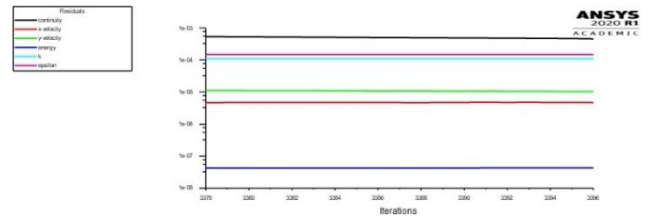


Figure5: residuals for 10s case1

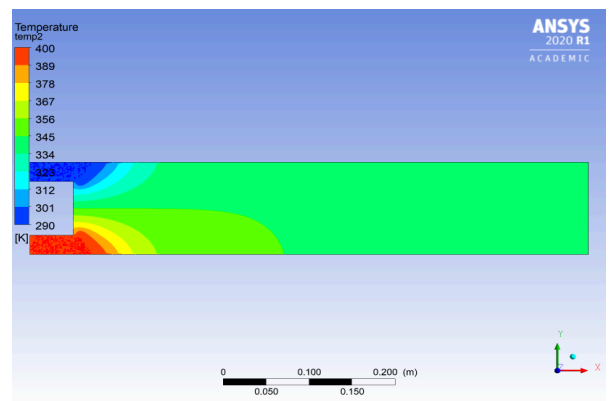


Figure6: temperature contour for 10 s case1

- time steps = 1000 flow time = 15s

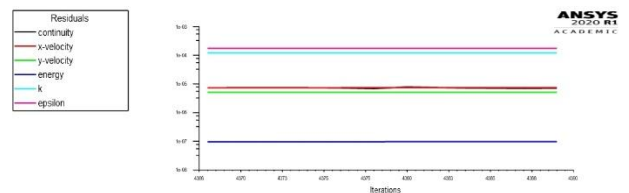


Figure7: residual for 15 s case1

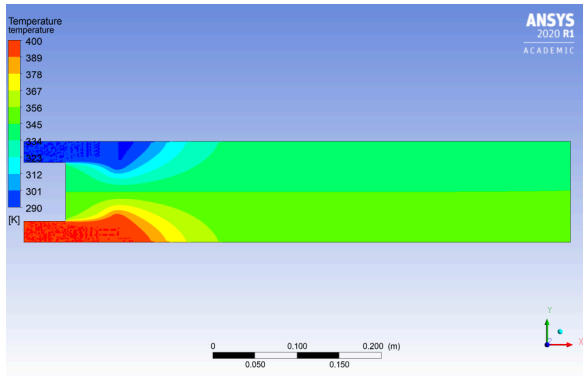


Figure8: temperature contour for 15s case1

- Time steps= 3000 flow time = 30s

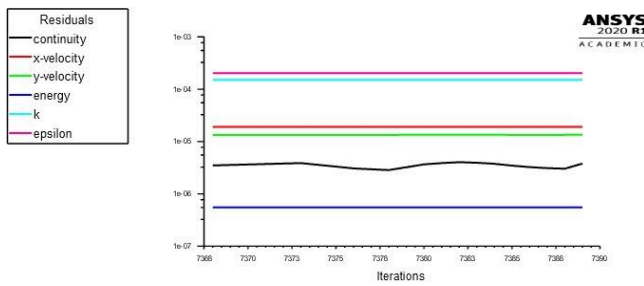


Figure9: residual for 30 s case1

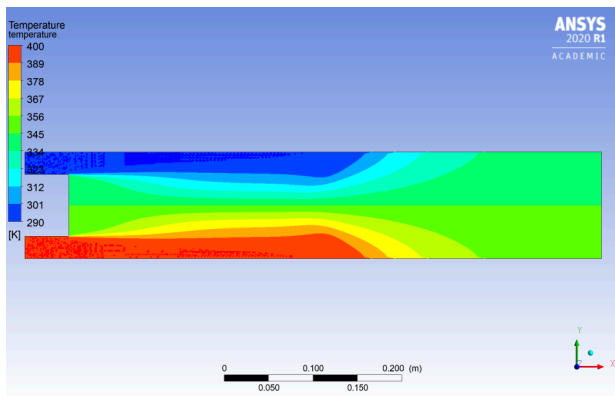


Figure10: temperature contour for 30s case1

- Time steps= 3000 flow time = 60s

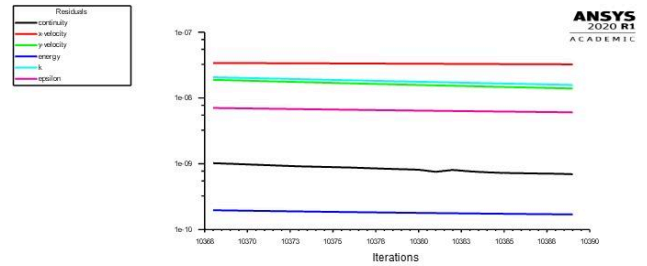


Figure11: residual for 60 s case1

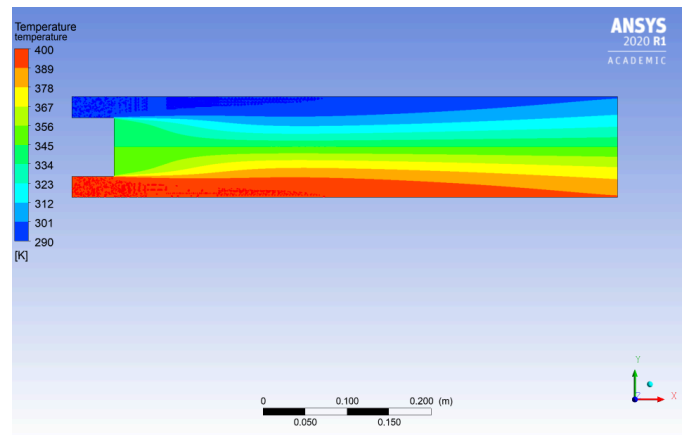


Figure12: temperature contour for 60s case 1

Temperature chart:

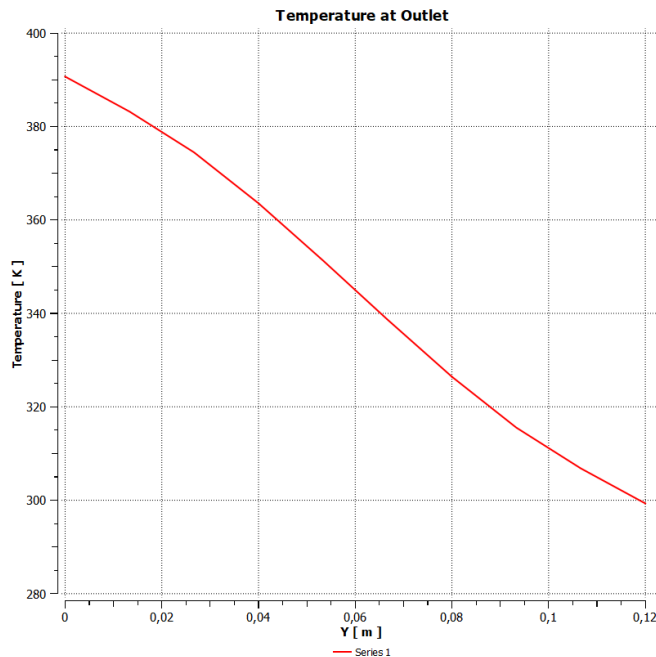


Figure13: outlet temperature chart for case1

Comment: In the temperature chart at the outlet temperature is decrease with increase Y. It should be line it means there is not mixing process at the outlet inside the chamber. The temperature does not distribute all of the outlet region. Mixing is not perfect.

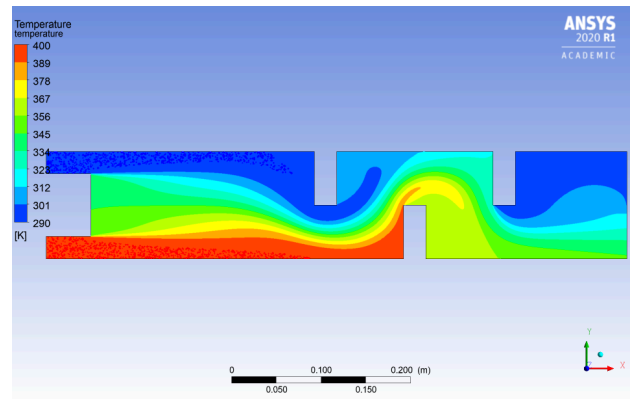


Figure15: temperature contour for 55s case2

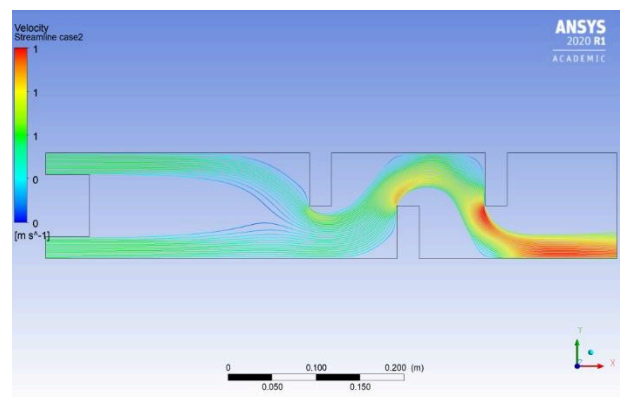


Figure16: streamline for 55s case2 at cold fluid inlet and hot fluid inlet

Comment: The resulting flow is concentrated at the bottom due to the vortex formed at the exit of the system. the incoming flow rate is not equal to the flow rate.

$$m_{in} < m_{out}$$

Residuals and temperature for case2:

- Timesteps = 11000 flow time: 55s

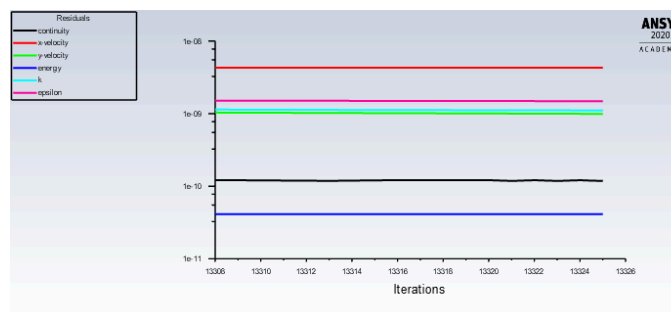


Figure14: residual for 55s case2

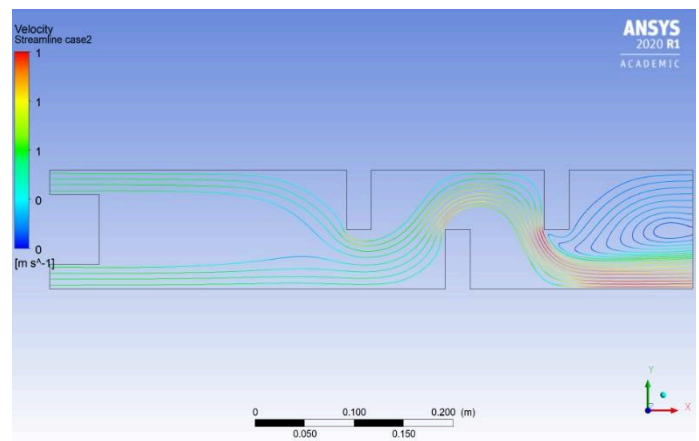


Figure17: streamline for 55s case2 at outlet (backflow)

Comment: the water entering the system can be seen to ensure the law of conservation of the mass. This flow is called backflow.

$$m_{in} = m_{out}$$

$$m_{in} + m_{flowback} = m_{out}$$

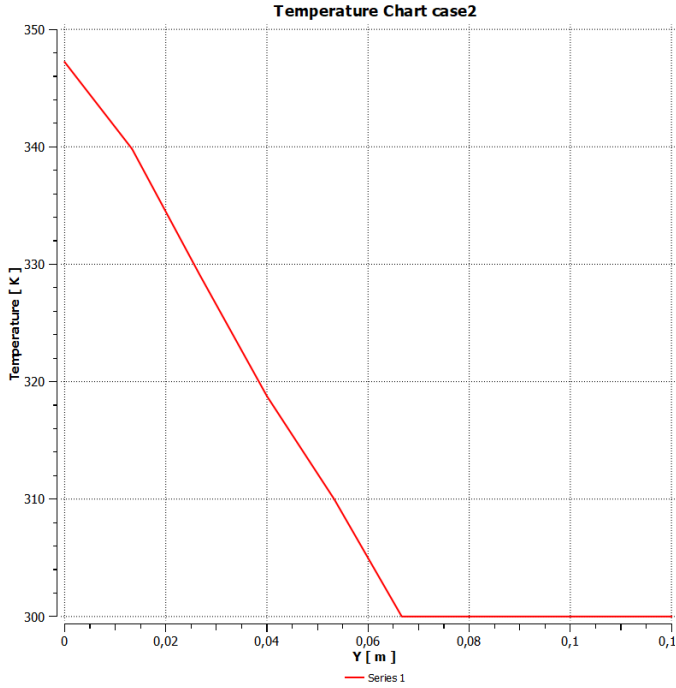


Figure18: outlet temperature chart for 55s case2

Comment: As the y axis decreases at the outlet, it is observed that the temperature increases and stabilizes at 300 K reflux.

4. CONCLUSION

To exemplify this analysis with a simple system; When we pour the jug, air comes in from the top of the jug while water comes out from the bottom. in our model, we modeled water as fluid. Since the program will only analyze with water, the program poured water into the mechanism from the top that was empty. This is due to the conservation of mass equation. The fluid exits by concentrating at the bottom of the outlet of the system due to the vortex formed after the last block. Vortices are better seen with the streamline contour. The output flow is larger than the incoming flow, so the program puts the fluid in the system as much as the difference between them. The values of this fluid are in the backflow section. We calculated the temperature of this water as 300K.

To reduce the backflow effect, we could define the outlet where the flowing fluid was dense.

REFERENCES

- [1] and [2] www.researchgate.net
- [3] and figure1: <https://scholarworks.uark.edu/>
- [4] <https://uotechnology.edu.iq/>